

Time Period	Variable	Label	Values	Definition	Group	Additional Notes
INDEX	INDEX_DATE	Index Date	date	date of the first medical claim for multiple myeloma	Tab 42	At least one inpatient medical claim with a diagnosis code for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) OR ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) in any position on claim, on separate days within 30/60/90 days. (Suggest to program this as a macro, where 30, 60 & 90 day windows can be applied to the outpatient rule). The index date is the date of the first medical claim for multiple myeloma. Diagnosis codes for MM are in tab 42. "MM dx code".
INDEX	INDEX_YR	Index Year	date	year(index_date)	Tab 42	
ID_PD	AGE_DX	Age year of index	numeric	year(index_date) - birthyear	N/A	
BASELINE	CE_b	Baseline Continuous Enrollment	Binary	if six months of CE with medical and pharmacy benefits before the index date then =1, else =0,	N/A	Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.
FU_PD	CE_f	Follow-up Continuous Enrollment of at least 1 day	Binary	Only 1 day of CE is required in the follow up period then =1, else =0,	N/A	Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled
FU_PD	CE_3mosf	Follow-up Continuous Enrollment of at least 3mos	Binary	Please flag patients if ≥ 3 months of CE starting on the index date and ending on the earliest of death, health plan disenrollment, or the study period end for patients who do not have an assigned date of death < 3 months after the index date then =1, else =0,	N/A	For patients with an assigned date of death < 3 months after the index date, the follow-up period ended at the assigned date of death. No gaps in enrollment are allowable. This patients may be subset on in the future. Patients should have no gaps in enrollment for continuously enrolled days that exceed 3mos.
FU_PD	MM_FU_agents	Evidence of treatment with agents for the purpose of treating MM	Binary	Evidence of an FDA-approved MM oncology therapy in claims during the follow-up period =1, else =0,	Tab 41	These will be identified with pharmacy claims (NDC) and medical claims (CPTs and HCPCS) codes.
BASELINE	MM_bl_agents	Evidence of MM oncology therapy in claims during the baseline period	Binary	≥ 1 medical or pharmacy claim for FDA-approved MM oncology therapy during the baseline period =1, else =0,	Tab 41	

	BASELINE	MM_baseline_diag	Evidence of any MM during the baseline period	Binary	Evidence of any MM (each of newly diagnosed and R/R MM), during the baseline period =1, else =0;	Tab 42 & 49	From Optum: "When we refer to a non-diagnostic multiple myeloma (or any disease) claim, what we are referring to is a claim where one of the multiple myeloma diagnosis codes is present, but there is at least one service line on the claim that doesn't equal a diagnostic code. The idea is that a physician/facility might order labs for testing/diagnostics for multiple myeloma and on the claim they have to put something for the diagnosis so they will put what they ordered the lab to test for. e.g., diagnosis code for multiple myeloma. If the resulting facility claim had a diagnosis code for multiple myeloma and only one service line on the claim and that service line indicated a laboratory test, then we can't be certain that this is a true multiple myeloma diagnosis because the claim is suggesting that only testing was done. Alternatively, if the claim had the diagnosis codes for multiple myeloma and two service lines: the first line for testing and the second line a HCPCS code for medication or a CPT code for physician care management then we will accept that claim as a "non-diagnostic claim" indicating multiple myeloma."	
14								
	BASELINE	MM_baseline_other	Evidence of another cancer in the baseline period	Binary	Evidence of another cancer in the baseline period =1, else =0;	Tab 44	Patients with ≥ 2 ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer. Only the first of the 2 codes is required to occur inside the baseline period.	
15								
16	ID_PD	Pregnant	Evidence of pregnancy	Binary	Evidence of pregnancy =1; else =0;	Tab 45	≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the patient identification period.	
17	ID_PD	CT_part	Clinical trial participation	Binary	Clinical trial participation =1, else =0;	Tab 46	Evidence of clinical trial participation during the patient identification will be flagged.	
	FU_PD	Death_dt	Death Date	Date	Date patient died.		When date of death is only available at month-level granularity, the date will be generalized to the middle of the month or year, respectively (e.g., a date of death in June 2014 will be treated as 15 June 2014). For patients with only death year available, the death date is assumed to be July 15th year of the year they died. For patients with only death year available and their year of death is the same year as their index date, the death	

A	B	C	D	E	F	G	H
5	INCLUSION		Flag patients with ≥ 6 months of Continuous Enrollment with medical and pharmacy benefits before the index date. The baseline period is fixed and is the six months before the index date. Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.				
34							
6	INCLUSION		Flag patients with ≥ 1 day of Continuous Enrollment starting on the index date. Follow up period ends on the earliest of death or the study period end.		Patients with ≥3 months of CE from the index date (with no allowable gaps in enrollment) will be flagged for consideration for sensitivity or secondary analyses, and ending on the earliest of health plan disenrollment or the study period for patients who do not have have an assigned date of death <3mos after the index date. For patients with an assigned date of death < 3 months after the index date, these patients are retained and have a limited follow-up period that ends at the assigned date of death.		
35							
7	EXCLUSION		Patients with ≥ 2 non-diagnostic diagnosis codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer		This is evidence of another cancer in the baseline period. Only the first of the 2 codes is required to occur inside the baseline period.		
37							
8	EXCLUSION		≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the baseline or follow-up period. See tab CL PREG.				
38							
9	EXCLUSION		Evidence of clinical trial participation during each of the baseline and follow-up periods. See tab CL CLINTRAL.				
39							

Section C: Final Study Population

Dataset	Label	Definition	Additional Notes	Date Modified (Initials + Date)
42				
ELIG_COH	A patient level dataset containing those patients for whom we would like to run the multiple myeloma LOT algorithm on. It is used as an input into the multiple myeloma LOT algorithm.	The team will begin with the requirement that patients have at least one inpatient medical claim with a diagnosis code for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) OR ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203.x or ICD-10-CM code=C90.x) in any position on claim, on separate days within 90 days during identification period. The team will apply all other IE criteria <u>with the exception</u> of the following exclusion criteria 1) Evidence of another cancer (criteria 7) 2) Pregnancy or childbirth during the baseline or follow-up period (criteria 8) 3) Evidence of RCT participation during the baseline and/or follow-up periods (criteria 9) *Patients that meet exclusion criteria 1-3 indicated here will be included in this build of the algorithm.	Apply each criteria as a flag that can be carried forward or merged onto a later dataset. Helps make the cohort dynamic and flexible to change future study requirements. Suggest to establish global variables upfront when creating inputs for the study population, or other approaches to make the initial inputs easy to change. Suggest to implement IE criteria using a function where follow up period, allowable gaps, etc. can be modified. There is only 1 day of CE required in the follow up period for this initial cohort build. Patients with ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) in any position on claim, on separate days within 30 and 60 days during identification period will be flagged in the attrition table as well. ELIG_COH dataset should include patient ID, index date, death date, study end date, number of follow up days, and all IE criteria flags should be retained.	
43				
44				
45				
46				
47				
48				
49				